

# Customer Shopping Behavior Intelligence

End-to-End Data Analytics Project | Python • SQL • Power BI

## 1. Project Context

In highly competitive retail environments, understanding what drives customer spending, loyalty, and product performance is critical.

This project analyzes 3,900 customer transactions to uncover behavioral insights and generate data-driven commercial strategies.

## 2. Business Objectives

- Identify high revenue customer segments
- Evaluate discount effectiveness
- Analyze subscription value contribution
- Understand demographic spending patterns
- Detect product pricing dependency

## 3. Dataset Overview

- Transactions: 3,900
- Features: 18
- Categories: Accessories, Clothing, Footwear, Outerwear

## 4. Data Preparation & Feature Engineering (Python)

- Data cleaning and preprocessing were performed using Pandas to ensure data reliability .
- Imputed missing review ratings using the median rating per product category.
- Renamed columns into snake case format for consistency.
- Created age group variable through age binning.
- Derived purchase frequency days metric.
- Removed redundant column promocode used.
- Cleaned dataset was loaded into MySQL for structured querying.

```
[23]: # Loading the dataset using pandas
```

```
import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
```

```
[24]: df.head()
```

```
[24]:
```

|   | Customer ID | Age | Gender | Item Purchased | Category | Purchase Amount (USD) | Location      | Size | Color     | Season | Review Rating | Subscription Status | Shipping Type | Discount Applied | Promo Code Used | Previous Purchases | Payment Method |
|---|-------------|-----|--------|----------------|----------|-----------------------|---------------|------|-----------|--------|---------------|---------------------|---------------|------------------|-----------------|--------------------|----------------|
| 0 | 1           | 55  | Male   | Blouse         | Clothing | 53                    | Kentucky      | L    | Gray      | Winter | 3.1           | Yes                 | Express       | Yes              | Yes             | 14                 | V              |
| 1 | 2           | 19  | Male   | Sweater        | Clothing | 64                    | Maine         | L    | Maroon    | Winter | 3.1           | Yes                 | Express       | Yes              | Yes             | 2                  |                |
| 2 | 3           | 50  | Male   | Jeans          | Clothing | 73                    | Massachusetts | S    | Maroon    | Spring | 3.1           | Yes                 | Free Shipping | Yes              | Yes             | 23                 |                |
| 3 | 4           | 21  | Male   | Sandals        | Footwear | 90                    | Rhode Island  | M    | Maroon    | Spring | 3.5           | Yes                 | Next Day Air  | Yes              | Yes             | 49                 | I              |
| 4 | 5           | 45  | Male   | Blouse         | Clothing | 49                    | Oregon        | M    | Turquoise | Spring | 2.7           | Yes                 | Free Shipping | Yes              | Yes             | 31                 | I              |

```
[ ]:
```

```
[28]: # Imputing missing values in Review Rating column with the median rating of the product category
```

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
[29]: df.isnull().sum()
```

```
[29]: Customer ID      0
Age                0
Gender             0
Item Purchased     0
Category           0
Purchase Amount (USD) 0
Location           0
Size              0
Color             0
Season            0
Review Rating      0
Subscription Status 0
Shipping Type      0
Discount Applied   0
Promo Code Used    0
Previous Purchases 0
Payment Method     0
Frequency of Purchases 0
dtype: int64
```

```
[30]: # Renaming columns according to snake casing for better readability and documentation
```

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
[31]: df.columns
```

```
[31]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

```
[34]: # create new column purchase_frequency_days

frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)

[35]: df[['discount_applied', 'promo_code_used']].head(10)
```

```
[35]:
```

|   | discount_applied | promo_code_used |
|---|------------------|-----------------|
| 0 | Yes              | Yes             |
| 1 | Yes              | Yes             |
| 2 | Yes              | Yes             |
| 3 | Yes              | Yes             |
| 4 | Yes              | Yes             |
| 5 | Yes              | Yes             |
| 6 | Yes              | Yes             |
| 7 | Yes              | Yes             |
| 8 | Yes              | Yes             |
| 9 | Yes              | Yes             |

```
[36]: (df['discount_applied'] == df['promo_code_used']).all()
```

```
[36]: np.True_
```

```
[37]: # Dropping promo code used column

df = df.drop('promo_code_used', axis=1)
```

## 5. Data Preparation & Feature Engineering

- Median imputation for missing ratings
- Created age cohorts and purchase frequency metrics
- Customer lifecycle segmentation (New / Returning / Loyal)
- MySQL integration for scalable querying

## 6. Key Insights

- Young Adults are the highest revenue contributors
- Express shipping users show higher purchase intent
- Certain products show ~50% discount dependency
- Loyal customers dominate the customer base
- Non-subscribers generate higher total revenue due to population size

## 7. Business Analysis Using SQL

- **Revenue Contribution by Gender** - Male customers generated significantly higher total revenue compared to female customers

| Result Grid |        |         |
|-------------|--------|---------|
|             | gender | revenue |
| ▶           | Male   | 157890  |
|             | Female | 75191   |

### High-Spending Discount Users

- **High-Spending Discount Users** - 839 customers were identified who used discounts but still spent above average.

| Result Grid |             |                 |
|-------------|-------------|-----------------|
|             | customer_id | purchase_amount |
|             | 1652        | 80              |
|             | 1654        | 93              |
|             | 1656        | 81              |
|             | 1659        | 66              |
|             | 1662        | 86              |
|             | 1667        | 64              |
|             | 1671        | 73              |
|             | 1673        | 73              |
|             | 1674        | 62              |
|             | 1676        | 90              |

- **Top 5 Products by Rating** – Found products with the highest average review ratings.

|   | item_purchased | avg review rating |
|---|----------------|-------------------|
| ▶ | Gloves         | 3.86              |
|   | Sandals        | 3.84              |
|   | Boots          | 3.82              |
|   | Hat            | 3.8               |
|   | Skirt          | 3.78              |

- **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

| Result Grid |                     |                 |           |               |
|-------------|---------------------|-----------------|-----------|---------------|
|             | subscription_status | total_customers | avg_spend | total_revenue |
| ▶           | Yes                 | 1053            | 59.49     | 62645         |
|             | No                  | 2847            | 59.87     | 170436        |

- **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

|   | item_purchased | discount_rate |
|---|----------------|---------------|
| ▶ | Hat            | 50.00         |
|   | Sneakers       | 49.66         |
|   | Coat           | 49.07         |
|   | Sweater        | 48.17         |
|   | Pants          | 47.37         |

- **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

|             |               |                                                                                   |                                                                                   |              |                      |   |
|-------------|---------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------|----------------------|---|
| Result Grid |               |  |  | Filter Rows: | <input type="text"/> | E |
|             | shipping_type | ROUND(AVG(purchase_amount),2)                                                     |                                                                                   |              |                      |   |
| ▶           | Express       | 60.48                                                                             |                                                                                   |              |                      |   |
|             | Standard      | 58.46                                                                             |                                                                                   |              |                      |   |

- **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

|   | customer_segment | number_of_customers |
|---|------------------|---------------------|
| ▶ | Loyal            | 3116                |
|   | Returning        | 701                 |
|   | New              | 83                  |

- **Top 3 Products per Category** – Listed the most purchased products within each category.

|   | item_rank | category    | item_purchased | total_orders |
|---|-----------|-------------|----------------|--------------|
| ▶ | 1         | Accessories | Jewelry        | 171          |
|   | 2         | Accessories | Sunglasses     | 161          |
|   | 3         | Accessories | Belt           | 161          |
|   | 1         | Clothing    | Blouse         | 171          |
|   | 2         | Clothing    | Pants          | 171          |
|   | 3         | Clothing    | Shirt          | 169          |
|   | 1         | Footwear    | Sandals        | 160          |
|   | 2         | Footwear    | Shoes          | 150          |
|   | 3         | Footwear    | Sneakers       | 149          |
|   | 1         | Outerwear   | Jacket         | 163          |
|   | 2         | Outerwear   | Coat           | 161          |

- **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

|   | subscription_status | repeat_buyers |
|---|---------------------|---------------|
| ► | Yes                 | 958           |
|   | No                  | 2518          |

- **Revenue by Age Group** – Calculated total revenue contribution of each age group.

| Result Grid |             | Filter Rows:  |
|-------------|-------------|---------------|
|             | age_group   | total_revenue |
| ►           | Young Adult | 62143         |
|             | Middle-aged | 59197         |
|             | Adult       | 55978         |
|             | Senior      | 55763         |

## 8. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



## 9. Strategic Recommendations

- Improve subscription benefits to increase CLV
- Optimize discount allocation using product elasticity signals
- Target high-value demographics for marketing ROI uplift
- Promote top-rated products for conversion improvement

- Strengthen loyalty incentives for retention growth

## **10. Project Takeaways**

This project demonstrates strong capability in data cleaning, SQL analytics, dashboard storytelling, KPI interpretation, and business strategy translation.